



FINAL PROJECT WORK
MASTER OF QUANTITATIVE FINANCE

Equity Factor allocation using a two-stage ML approach

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Embracing the stillness of a new dawn.

Abstract

The rapid advancement of machine learning (ML) has significantly transformed various industries, with finance being one of the most impacted sectors. As financial markets become increasingly complex and data-driven, the adoption of ML techniques offers powerful tools for predictive analytics, risk management, and investment strategy optimization. This project builds on a recent study on the timing of equity factors using a two-stage machine learning framework [1], highlighting the growing incorporation of ML into financial decision-making processes.

This study investigates a method to enhance the predictability of equity factor performance under varying market conditions. The first stage of the model utilizes both unsupervised and supervised machine learning to identify dynamic market risk regimes, incorporating various macroeconomic indicators and market risk signals as the Financial Turbulence. In the second stage, supervised ensemble learning techniques are applied to predict factors importance within each identified regime. By dynamically adjusting factor exposure based on these predictions, the model seeks to outperform traditional static factor investment strategies. The inclusion of advanced ML methodologies demonstrates the potential for achieving superior investment performance and managing risk more effectively, illustrating the continuing role of machine learning in shaping the future of financial markets.

The results of this work demonstrate that the machine learning-based factor timing approach can outperform traditional static investment strategies, particularly in stable market conditions.

Keywords: machine learning, equity factor timing, stacking classification approach, financial turbulence

Contents

Abstract	i
Contents	iii
Introduction	1
1 Market risk regime analysis	3
1.1 Part 1: Unsupervised learning	3
1.1.1 SP500 three months' rolling-drawdown calculation	3
1.1.2 K-means clustering	4
1.1.2.1 Theoretical recalls	4
1.1.2.2 Outcomes	6
1.2 Part 2: Supervised learning	7
1.2.1 Analysis of macroeconomic indicators	7
1.2.2 Calculation of the financial turbulence	8
1.2.3 Implementation of the classification models	9
1.2.3.1 Why wasn't it feasible with 3 market regimes (classes)? . .	9
1.2.3.2 Classification models with the 2 dominant market regimes	12
2 Factors analysis	15
2.1 Introduction to market factors	15
2.2 Correlation between factors	17
2.3 Random Forest classification models	18
2.3.1 Preprocessing Data	18
2.3.2 Digging into classification models	19
3 Strategies results	22
3.1 Comparison between strategies	23
3.2 Long-short strategies	27

4 Conclusion	29
Bibliography	31
A Appendix A	32
B Appendix B: Features selection	34
C Appendix C: Overview of classification model parameters	36
List of Figures	39
List of Tables	40

Introduction

Over the past decades, factor investing has increasingly drawn the attention of the finance community, transforming traditional investment strategies. The work of Fama and French (1992) [2] introduced a multifactor approach to investing, extending Sharpe's Capital Asset Pricing Model (CAPM)[7]. They demonstrated that by focusing on specific factors influencing asset pricing and performance, investors could potentially achieve enhanced returns. As factor investing has evolved, key elements that shape equity investment strategies have been identified. These factors include size, value, growth, quality, momentum, and leverage. By examining these factors, investors can develop strategies aligned with their specific investment goals, as demonstrated by Fama and French[3].

According to Kevin J. DiCiurcio et al. [1], only a limited number of studies have thoroughly explored the effects of variations in factor investments, whether and how tactical allocation strategies can boost risk-adjusted returns and improve the overall performance of factor-based portfolios. This gap offers a valuable opportunity for further enhancement of strategies based on investment factors.

In addition, it should be emphasised how prior research has revealed a strong link between macroeconomic elements and the relative performance of various investment factors. In their study, Kao and Shumaker [4] demonstrated that certain factors outperform others due to specific economic fundamentals and stages of the business cycle. In 2020, Polk, Haghbin, and de Longis [6] offered significant insights into how factors behave cyclically by interpreting signals that predict shifts in the business cycle from a factor perspective. They examined five well-known factors: low volatility, size, value, momentum, and quality. By adjusting the weighting of these factors during various phases of the business cycle - such as emphasizing size and value during recovery periods, prioritizing momentum in expansion phases, and focusing on low volatility and quality during slowdowns and contractions - they implemented a factor timing strategy based on the business cycle. This approach led to performance that surpassed their benchmark, the Russell 1000 Index, highlighting the practical benefits of integrating macroeconomic views and cyclical perspectives into factor-based investment strategies.

Relying exclusively on the business cycle to time investment factors can oversimplify the complex dynamics of financial markets. Moreover, macroeconomic indicators often change gradually, leading to delays in adjusting to sudden market shifts. To address these challenges, Kevin J. DiCiurcio et al. [1] introduces a machine learning approach to enhance factor timing by integrating macroeconomic variables and additional risk factors into the model. Specifically, the study utilizes the financial turbulence ratio, a measure of systemic risk identified by Kritzman and Li (2010) [5]. By incorporating these elements within a machine learning framework, the aim is to improve the precision and adaptability of factor timing methods, enabling investors to make more informed decisions.

This project builds on the work of Kevin J. DiCiurcio et al [1]. However, while it was initially possible to closely follow the methodology outlined in the reference paper, at a certain point, different choices had to be made and alternative data were used. As a result, the outcomes are no longer directly comparable. This report is structured to present the process in a step-by-step manner, allowing each phase to be followed clearly. Relevant theoretical concepts will be introduced as new topics are addressed throughout the discussion.

Chapter 1 will be devoted to the analysis of market regimes, followed in Chapter 2 by the factor analysis. Chapter 3 will present the results followed by conclusions in Chapter 4. Let's delve into it.

1 | Market risk regime analysis

In the first part, the goal is to categorize market risk regimes using K-means clustering on SP 500 maximum drawdown data. This clustering helps identify and label distinct groups that reflect varying market conditions.

Following this, a supervised approach is employed to predict these regimes using the labels generated from the previous step. Feature selection based on a random forest approach or mutual information could have been useful at this stage if the amount of features available was too great. At first, however, it was preferred to proceed by setting this aside for later. Addressing the issue of an imbalanced classes was crucial here as some market regimes were underrepresented. To mitigate potential model bias resulting from this imbalance the Synthetic Minority Over-sampling Technique (SMOTE) was implemented. However, quite a few problems arose at this stage, which led to acting accordingly, departing not a little from the approach used in the reference paper. More details on the procedure are presented in the following paragraphs.

1.1. Part 1: Unsupervised learning

1.1.1. SP500 three months' rolling-drawdown calculation

In order to identify market regimes, first, it was necessary to calculate and visualizes the SP500's drawdown based on daily returns, focusing on a three-month rolling drawdown and resampling the data to a monthly level. The SP500 is used for to represent market regimes because it offers broad market representation, covering around 80% of U.S. stock market capitalization and because its long historical record allows analysis of various market cycles making it an effective indicator of market shifts. Additionally, the SP500's global influence and role as a benchmark for investors further solidify its suitability for analyzing drawdowns and market regimes.

Initially, daily returns are calculated from the SP500 closing prices. Then, a rolling maximum value of the cumulative returns over the past 63 trading days (approximately three months) is computed. The drawdown is derived as the difference between the

cumulative returns and the rolling maximum, normalized by the rolling maximum value. It should be specified how the **maximum** drawdowns for each month were taken into account in the resampling.

The figure 1.1 represents the monthly drawdown from 1987 to 2023.

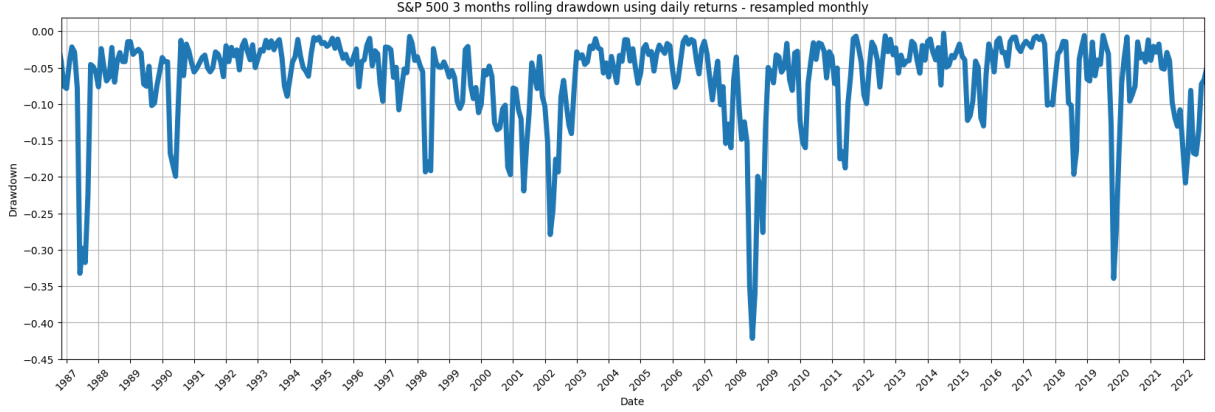


Figure 1.1: S&P500 monthly drawdown

1.1.2. K-means clustering

1.1.2.1. Theoretical recalls

K-Means clustering is an unsupervised learning algorithm used to partition data into k clusters based on similarities. Each observation belongs to the cluster with the nearest mean (centroid), minimizing intra-cluster variance. The objective of K-Means is to minimize the within-cluster sum of squares (WCSS), also called inertia, which is the sum of squared distances between data points and their assigned cluster centroids.

Mathematical formulation of K-Means: Let $X = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ be the set of observations, where each $\mathbf{x}_j$ is a vector in a multi-dimensional feature space, and let $C_1, C_2, \dots, C_k$ represent the clusters. The goal of K-Means is to minimize the following objective function:

$$\text{Inertia} = \sum_{i=1}^k \sum_{\mathbf{x}_j \in C_i} \|\mathbf{x}_j - \boldsymbol{\mu}_i\|^2$$

where:

- $\mathbf{x}_j$ is a vector representing the j -th observation (data point),
- $\boldsymbol{\mu}_i$ is the **centroid vector** (mean) of the points in cluster C_i ,

- $\|\mathbf{x}_j - \boldsymbol{\mu}_i\|^2$ is the squared Euclidean distance between the vector $\mathbf{x}_j$ and the centroid $\boldsymbol{\mu}_i$,
- k is the number of clusters.

The squared Euclidean distance $\|\mathbf{x}_j - \boldsymbol{\mu}_i\|^2$ is calculated as:

$$\|\mathbf{x}_j - \boldsymbol{\mu}_i\|^2 = \sum_{d=1}^p (x_j^{(d)} - \mu_i^{(d)})^2$$

where p is the number of features (dimensions), $x_j^{(d)}$ is the d -th feature of observation $\mathbf{x}_j$, and $\mu_i^{(d)}$ is the d -th feature of the centroid $\boldsymbol{\mu}_i$. In the case under examination, only one feature is represented by drawdowns, so the calculation is simplified.

The algorithm proceeds iteratively by:

1. **Assigning** each point to the nearest centroid.
2. **Updating** each centroid to be the mean of the points assigned to it.

The iterations stop when the centroids no longer change, or the algorithm reaches a predefined number of iterations.

Choosing the optimal number of clusters: The Elbow Method.

The number of clusters k must be predefined. The Elbow Method is used to determine the optimal number of clusters. The method involves plotting inertia for different values of k and identifying visually the "elbow" point, where the rate of inertia reduction slows down. The elbow point represents the optimal balance between the number of clusters and the clustering quality. In this case, the Elbow Method is suitable for identifying distinct market regimes because it ensures that additional clusters beyond the elbow point do not significantly reduce inertia, highlighting key market behaviors.

Looking at the Figure 1.2 and also following what was done in the paper [1], it was at first decided to continue with $k = 3$. A posteriori, it became clear that 3 market regimes greatly complicated the discussion in the following section 1.2. More details on this will be given later.

1.1.2.2. Outcomes

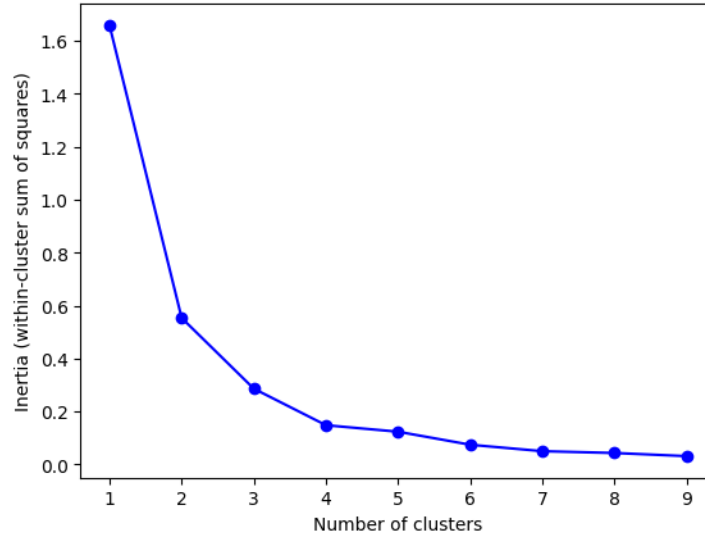


Figure 1.2: Elbow method visualization

The table below presents a comparison between the results of the clustering analysis done and the findings from the referenced paper. The percentage of time spent in each regime is shown for both analyses, highlighting the differences and similarities in regime classification.

Market Regime	Result (% , Samples)	Paper Result (%)
Normal Risk Market Regime	77.37% (335)	77.5%
Correction Market Regime	19.86% (86)	18.7%
Bear Market Regime	2.77% (12)	3.8%

Table 1.1: Comparison of market regime clustering results

It is clear that the results are very close to each other, a symptom of the correct approach implemented so far. However, it is worth noting that for the minority regime, there are only 12 samples available, which led to quite a few problems in the next section and the consequent decision to consider only 2 of the 3 market regimes by including the bear regime in the correction one. A visualization of market regimes and drawdown together is shown below:

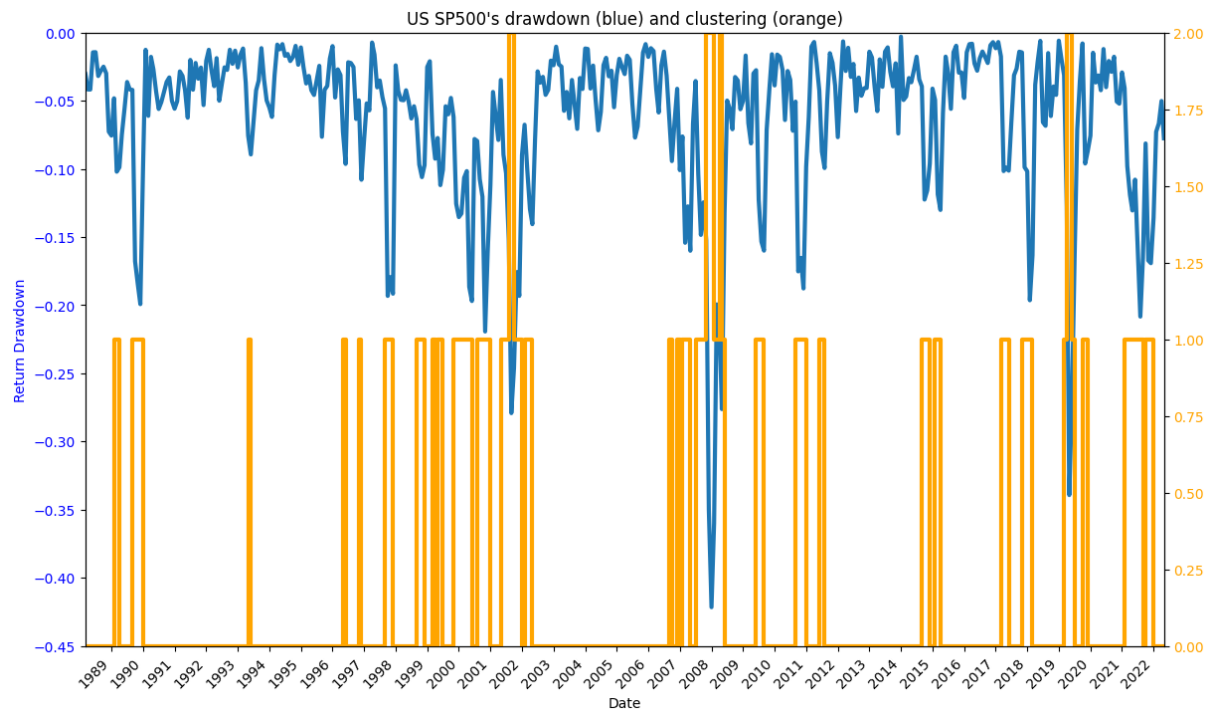


Figure 1.3: Clustering and drawdown visualization

1.2. Part 2: Supervised learning

This second part aims to train and test a classification model capable of, given certain input variables, correctly predicting the corresponding market regime. At first, the input variables considered will be dealt with, only afterwards the modelling part will be discussed in more detail.

1.2.1. Analysis of macroeconomic indicators

As already mentioned in the introduction, most of the variables that will be referred to are macroeconomic variables. These variables span a broad range of economic indicators, including price indices, employment statistics, financial conditions, and economic activity measures. Each variable plays a critical role in analyzing the state of the economy, making them essential for forecasting and economic modeling. Given the large number of macroeconomic variables in order not to slow down the reading, the summary table will be left as an annex A.1 Note how these variables are on different scales and/or with different units of measurement. The data for most of these were downloaded from the FRED website with the exception of CPILFESL1YAC and CPILFESL10YAC obtained from CPILFESL and CAPE downloaded from Shiller Data

1.2.2. Calculation of the financial turbulence

As mentioned in the introduction Kritzman and Li (2010) [5] introduced a novel method for detecting systemic risk in financial markets through the concept of financial turbulence. This concept describes situations where asset prices deviate significantly from their historical patterns, often marked by extreme price movements, the breakdown of correlations between previously linked assets, and the unexpected convergence of previously uncorrelated assets. Understanding financial turbulence provides valuable insights into market dynamics and helps in assessing the risks of broader market disruptions. This variable would serve to give an extra edge to the information contained in the macroeconomic variables alone.

Following their methodology, this report uses monthly returns from ten SP500 sectors (information technology, energy, financials, healthcare, consumer staples, consumer discretionary, utilities, industrials, telecommunications, and materials) along with returns from Treasury instruments with maturities of 12 months, 2 years, and 10 years to construct a financial turbulence index. This approach emphasizes that determining market risk regimes requires consideration of the evolving nature of market conditions, rather than focusing solely on current market states. The reference paper does not mention how the yields are calculated, which is why several tests were carried out (percentage yields, logarithmic, absolute, ...) to ensure that the results were as compatible as possible.

The financial turbulence for a given time period t is calculated as:

$$d_t = (\mathbf{y}_t - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{y}_t - \boldsymbol{\mu})$$

where:

- d_t = turbulence for a particular time period t
- $\mathbf{y}_t$ = vector of asset returns for period t
- $\boldsymbol{\mu}$ = sample average vector of historical returns
- $\boldsymbol{\Sigma}$ = sample covariance matrix of historical returns

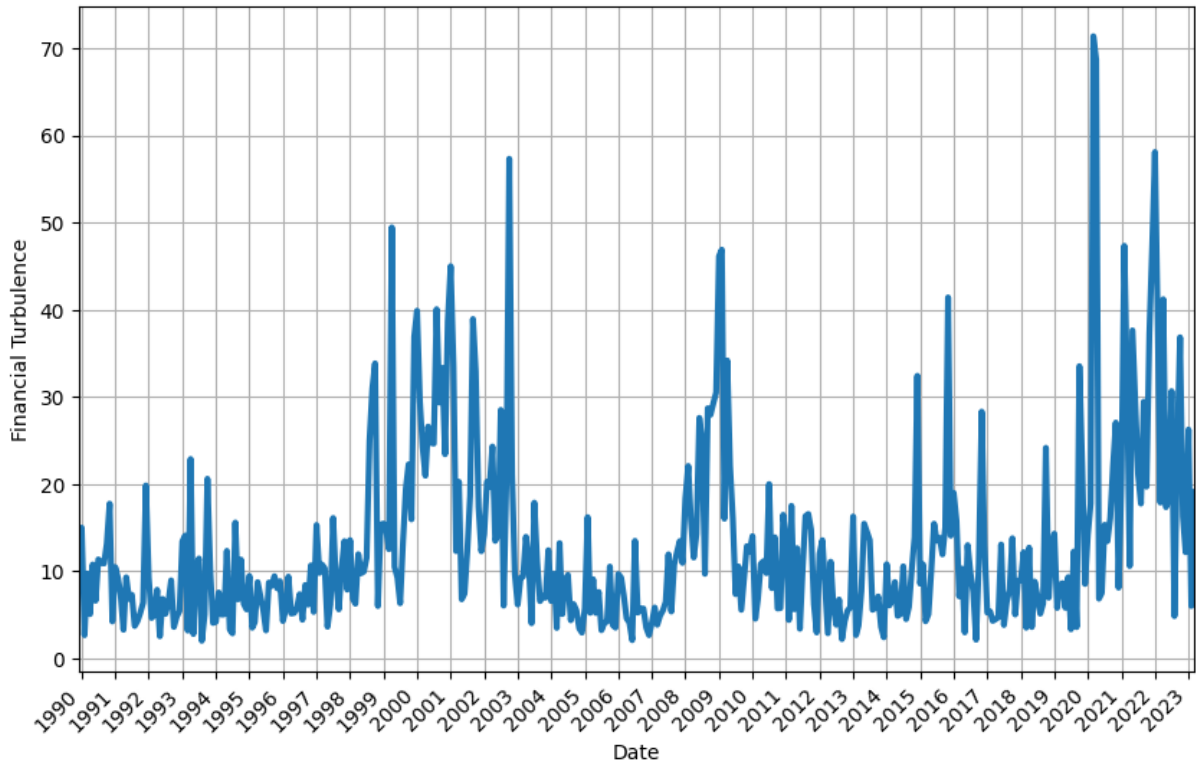


Figure 1.4: Financial turbulence

The figure 1.4 plots the value of financial turbulence from 12/1989 to 03/2023

Financial turbulence along with all the macroeconomic variables shown in Appendix A will be used to train the ML models presented below.

1.2.3. Implementation of the classification models

With all the necessary data available, the next step is to identify the model that best fits the problem in question in order to develop a binary or multiclass classification. A first attempt was made to use a three-class classification meant as 3 market regimes (normal, correction, bear) along the lines of the procedure described in the paper [1]. However, in doing so, more than a few difficulties arose. Below is a review of what has been done.

1.2.3.1. Why wasn't it feasible with 3 market regimes (classes)?

This first approach implemented a Stacking Ensemble Classification approach to enhance predictive performance, combining three base classifiers—Random Forest (RF), Support Vector Classifier (SVC), and Gaussian Naive Bayes (GNB)—with a Logistic Regression as the meta-classifier. The stacking model was optimized by learning the appropriate

weights assigned to the predictions of the base classifiers using cross-entropy loss. The primary benefit of the stacking classification approach compared to using individual base models lies in its ability to leverage the strengths of multiple models to create a more robust and generalized predictive system. From a more mathematical point of view, the idea behind it is as follows.

Mathematical formulation of the Stacking Ensemble Classifier: In a multi-class classification setting, the stacking ensemble produces a probability for each class. These probabilities are calculated by combining the outputs of the three base classifiers (Random Forest, Support Vector Classifier, and Gaussian Naive Bayes). For each class c , the model outputs a probability that class c is the correct classification, with the probabilities for all classes summing to 1.

$$L(w) = - \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(y_{\text{pred}_{i,c}})$$

Where:

- N is the number of samples,
- C is the number of classes (in this case, $C = 3$),
- $y_{i,c}$ is a binary indicator (0 or 1) if class c is the correct classification for sample i ,
- $y_{\text{pred}_{i,c}}$ is the predicted probability for class c for sample i ,
- w represents the 9 weights to be learned by the meta-classifier.

For each class c , the predicted probability for sample i , $y_{\text{pred}_{i,c}}$, is calculated by combining the probability-based outputs from the 3 base classifiers (Random Forest, Support Vector Classifier, and Gaussian Naive Bayes). The formula is:

$$y_{\text{pred}_{i,c}} = \sigma(w_{1,c} \cdot RF(x_i)_c + w_{2,c} \cdot SVC(x_i)_c + w_{3,c} \cdot GNB(x_i)_c)$$

Where:

- $RF(x_i)_c$ is the output from the Random Forest for class c ,
- $SVC(x_i)_c$ is the output from the Support Vector Classifier for class c ,
- $GNB(x_i)_c$ is the output from the Gaussian Naive Bayes for class c ,
- $w_{1,c}, w_{2,c}, w_{3,c}$ are the learned weights for each base classifier's output for class c .

The final predicted probability for each class is computed using the softmax function, which ensures that the sum of probabilities across all classes equals 1. The softmax function is defined as:

$$\sigma(z_i)_c = \frac{\exp(z_{i,c})}{\sum_{j=1}^C \exp(z_{i,j})}$$

Where:

- $z_{i,c} = w_{1,c} \cdot RF(x_i)_c + w_{2,c} \cdot SVC(x_i)_c + w_{3,c} \cdot GNB(x_i)_c$,
- C is the number of classes (in this case, $C = 3$).

The complete multi-class stacking model can now be written as:

$$L(w) = - \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(\sigma(w_{1,c} \cdot RF(x_i)_c + w_{2,c} \cdot SVC(x_i)_c + w_{3,c} \cdot GNB(x_i)_c))$$

Synthetic Minority Over-Sampling Technique (SMOTE): To address the class imbalance present in the dataset, particularly in the minority bear market regime, SMOTE (Synthetic Minority Over-Sampling Technique) was employed. SMOTE creates synthetic samples for the minority class by interpolating between existing data points, effectively augmenting the underrepresented class and improving classification performance.

Given a minority class sample x_i and one of its K nearest neighbors x_j , a new synthetic sample x_{new} is generated using the following formula:

$$x_{\text{new}} = x_i + \alpha \cdot (x_j - x_i)$$

Where:

- x_i is a sample from the minority class,
- x_j is one of the K nearest neighbors of x_i ,
- α is a random number between 0 and 1.

This formula creates new synthetic data points along the line segment between x_i and x_j , providing the model with additional diverse examples from the minority class and mitigating the class imbalance problem.

SMOTE was applied to the training data before feeding it to the base classifiers (Random Forest, SVC, and Gaussian Naive Bayes) in the stacking ensemble model.

However, challenges arose when dividing train-test data and/or applying SMOTE within the context of time series data during cross validation. Using an approach randomly splitting the data into training and testing sets (70%/30%) would have led to data leakage. However, even if this had not been a problem, the reduced number of samples in the minority class would have created quite a few limitations in the part related to cross validation. To avoid altering the training of the model, the alternative would therefore be to partition the training data respecting the temporal trend and this is where the problem arises. By doing so, in cross validation there are cases in which no samples of the minority class are present in the training folds. This negatively impacts the process, making it unsuitable for the case under consideration. This is why it was decided to proceed on only 2 market regimes by including the bear regime samples in the correction market regime.

1.2.3.2. Classification models with the 2 dominant market regimes

With only two classes, the problems highlighted above do not arise. While the previous section focused mainly on why that strategy could not be used, this section will go into more detail on the process that led to the final model used in phase 2. There will be a part on data preprocessing, followed by an introduction to feature selection, concluding with models analysis

Preprocessing data

Preprocessing is an important step in machine learning as it prepares raw data for model training, improving the quality and performance of the model. It involves tasks like normalizing or scaling features, encoding categorical variables, removing outliers, stationarising data.

Since the input data were handled directly, it was ensured that no procedures related to their absence and format were necessary. The pre-processing was divided into 2 phases: stationarisation and min-max scaling.

1. Stationarisation

Stationarisation is a crucial step in time series analysis to ensure that the statistical properties of the data, such as the mean and variance, remain constant over time. This is essential for many machine learning models, especially those that rely on temporal patterns. In this case, the Augmented Dickey-Fuller (ADF) test was employed to test for stationarity. The procedure was applied one or two times depending on the nature of the variable. In the first application the ADF test was run on each variable. If the test result indicated non-stationarity (i.e., the p-value

was above a predefined threshold, typically 0.05), a first-order differencing was performed to remove the trend. For some variables, after the first differencing, the ADF test still indicated non-stationarity. In such cases, a second differencing was applied to remove any remaining trend or seasonality in the data.

2. Min-Max scaling

Min-max scaling is applied to transform the features into a fixed range, typically between 0 and 1. The formula used for min-max scaling is:

$$X_{\text{scaled}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

Where:

- X is the original value of the feature,
- $X_{\min}$ is the minimum value of the feature in the dataset,
- $X_{\max}$ is the maximum value of the feature in the dataset.

Classification models

Trying to maintain a 70-30 ratio between training and test data, 01-01-2010 was chosen as the split date. The next step was to start testing the classification models. However, the initial aim was to get to observe the performance of the stacking ensemble approach without, however, disregarding the performance of the basic models. The random forest, support vector machine and Gaussian Naive Bayes models were analyzed individually. For each of these, an attempt was made to identify the optimal hyperparameters using Gridsearch (or alternatively RandomSearch if the process took too long) and the HyperOpt library. The analysis of the characteristic parameters for each model is found in Annex C C.

Results: This section will report the output parameters of the Random Forest and Stacking classification approach models showed in 1.2 and 1.3. The former resulted in better output and was chosen as the reference model. However, it can be seen that the recall for the class 1 is low. This is due to a tendency of the model to predict class 0 and is related to an imbalance between classes. Despite therefore acting using SMOTE oversampling to address this problem, it is still possible to improve this aspect.

Table 1.2: Test results for the RF model

Class	Precision	Recall	F1-score	Support
0	0.87	0.98	0.92	122
1	0.90	0.51	0.66	37
Accuracy	0.87			
Macro avg	0.89	0.75	0.79	159
Weighted avg	0.88	0.87	0.86	159

Table 1.3: Test results for the SCA

Class	Precision	Recall	F1-score	Support
0	0.85	0.91	0.88	122
1	0.61	0.46	0.52	37
Accuracy	0.81			
Macro avg	0.73	0.68	0.70	159
Weighted avg	0.79	0.81	0.80	159

2 | Factors analysis

In the second chapter, a factor timing strategy is developed using six common equity factors: value, growth, momentum, low volatility, quality, and small size. If the results from the market regime classification in the first part play a crucial role in informing this strategy by identifying whether the market is in a normal or correction regime, here the goal is to understand, through the use of machine learning algorithms, how to weight each factor in the investment portfolio taking into account the market regime.

2.1. Introduction to market factors

Historically each of these factors represents a distinct investment strategy and they are known to react differently under various market conditions. A brief description of these is presented below.

Value: It identifies stocks that are priced lower relative to their fundamental worth. These stocks are often considered undervalued by the market and are expected to perform better in the long run once their prices align with their intrinsic value. Common ways to measure the value factor include the price-to-earnings (P/E) ratio, the price-to-book (P/B) ratio.

Growth: The growth factor focuses on companies that are expected to grow faster than the market or their peers. Growth stocks are characterized by strong earnings and revenue growth. These companies often reinvest their profits into expansion rather than returning capital to shareholders. Growth stocks tend to perform well in bullish markets, and metrics such as earnings growth, revenue growth, and the price-to-earnings growth (PEG) ratio are commonly used to identify them.

Momentum: The momentum factor takes advantage of the tendency of stocks that have performed well in the recent past to continue performing well in the near future. This strategy relies on the behavioral bias of investors who tend to chase winners. Momentum is typically measured by looking at a stock's performance over the past 6 to 12 months. Stocks with high relative price strength are considered momentum stocks. The idea is to capitalize on continued upward trends.

Low Volatility: The low-volatility factor targets stocks that experience smaller fluctuations in price over time. These stocks are considered less risky and are often favored during market downturns when investors seek safety. Low volatility is typically measured using the stock's historical price volatility or beta, which reflects how much a stock moves in relation to the broader market. Stocks with lower beta values tend to be less volatile and are preferred in uncertain market conditions.

Quality: The quality factor highlights companies with strong financial fundamentals. These firms are generally profitable, have stable earnings, low debt, and efficient use of capital. Companies with high return on equity (ROE), consistent earnings, and low debt-to-equity ratios are considered high quality. Quality stocks are often more resilient during economic downturns and tend to provide consistent performance over time.

Small Size: The small-size factor reflects the historical tendency for smaller companies to outperform larger ones over the long term. Smaller firms, typically measured by market capitalization, are seen as having higher growth potential but also come with greater risk. Market capitalization is calculated by multiplying a company's stock price by the number of outstanding shares. Small-cap stocks are known to outperform in periods of economic expansion and during market recoveries.

Equity Factors	Definition
Value	Value stocks based on three factors: the ratios of book value, earnings, and sales to price.
Growth	Growth stocks based on three factors: sales growth, the ratio of earnings change to price, and momentum.
Momentum	Securities in the S&P 500 universe that exhibit persistence in their relative performance.
Low Volatility	Constituents are weighted relative to the inverse of their corresponding volatility, with the least volatile stocks receiving the highest weights.
Quality	High quality stocks in the S&P 500 by quality score calculated based on return on equity, accruals ratio, and financial leverage ratio.
Small-Cap	S&P The index (SmallCap 600) is designed to track companies that meet specific inclusion criteria to ensure that they are liquid and financially viable.

Table 2.1: Selected equity factors based on the S&P 500 Index

The factors used in this analysis are based on the **S&P 500 Index** (see table 2.1), in contrast to the **Russell 1000** and **Russell 3000** indices employed in the reference paper [1]. Unfortunately, it was not possible to retrieve the necessary data from the available platforms. This difference impacts of course the outcomes because the last two indices cover a broader spectrum of companies, including smaller-cap stocks, and provide a more comprehensive representation of the U.S. equity market. By focusing on the S&P 500, the strategy misses exposure to smaller companies that are part of the **Russell 3000**, especially those in the **Russell 2000**, which could lead to different results, particularly regarding the **Small-Cap** factor. Available data on SP500 factors range from 1996 to 2023

2.2. Correlation between factors

Correlation in factor-based investing is a double-edged sword: while it can help identify relationships between factors, high correlations can undermine diversification and complicate the process of factor selection. Using market regimes to break down the data into normal and correction periods provides a more dynamic framework for understanding how factors perform in different conditions. This approach often reduces correlation between factors within each regime, making it easier to identify and allocate to the outperforming factors depending on the prevailing market environment. Below a visualization of the results.

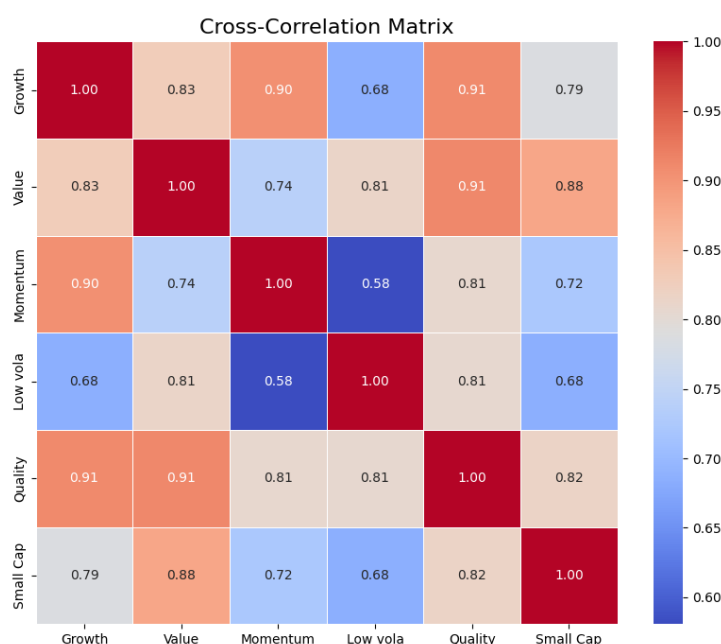


Figure 2.1: Global correlation matrix

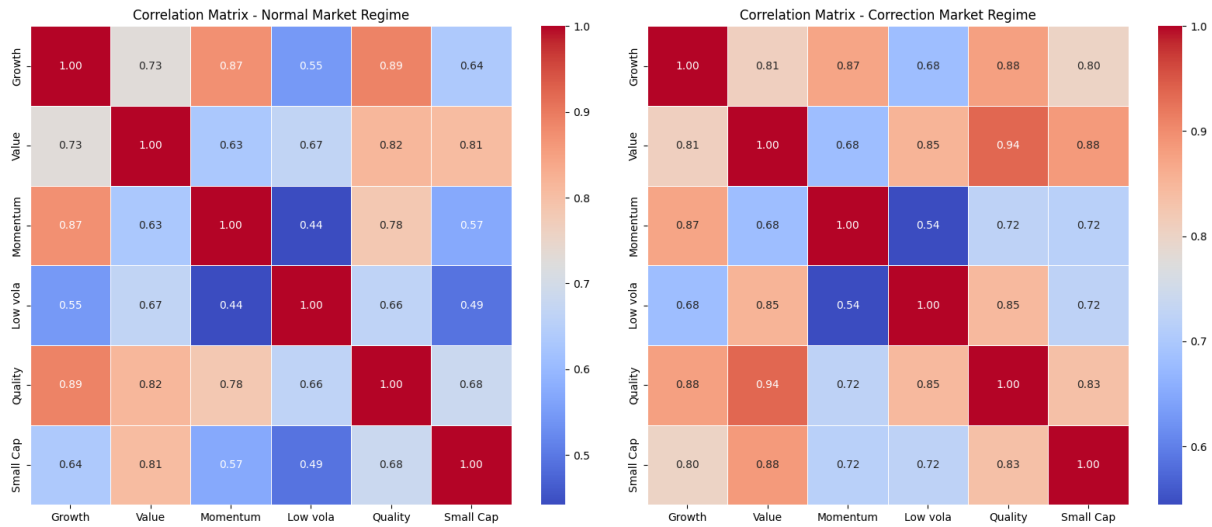


Figure 2.2: Normal market correlation matrix (left) and Correction market correlation matrix (right)

2.3. Random Forest classification models

2.3.1. Preprocessing Data

The part related to stationarisation and scaling follows what has already been seen in the first part. However, whereas in the first part the class labels were obtained from the output of the clustering, here it is necessary to act differently.

Definition of the target variable: The target variable was created using the one-hot encoding method, which is essential for identifying the factors that outperform in each market regime. This approach is particularly useful for a multi-class classification task such as this. Further details on the process will be provided in the following lines.

- **Step 1:** For each month, the performance of the factors (such as Growth, Value, Momentum, Low Volatility, Quality, and Small Cap) is evaluated. The factor with the highest return in a given month is identified as the *winning factor*, which becomes the label for that month, indicating that it outperformed the others.
- **Step 2:** Once the winning factor is identified, one-hot encoding is used to represent this information. One-hot encoding involves creating a binary vector for each month. A value of **1** is assigned to the position corresponding to the winning factor, while the other positions in the vector are assigned a value of **0**. This transformation turns the categorical label (the winning factor) into a format that a machine learning

model can process, indicating which factor outperformed while marking the others as non-winning.

The labeled data, now encoded with one-hot vectors for each month, is then used to train the machine learning model. The model learns to predict the probability of each factor outperforming others in future periods, aiding in making strategic decisions based on these predictions. On a historical basis over the time interval considered, the table 2.2 shows how the probability of winning is distributed for each factor.

Factor	Winning Probability (%)
Growth	8.59
Value	22.69
Momentum	19.02
Low Volatility	10.74
Quality	31.29
Small Cap	7.67

Table 2.2: Winning probabilities for each factor

Once the data labelling is complete, the modelling procedure follows a precise pattern. A random forest model for each market regime. Each model will allow once trained to predict the probability of outperformance of each factor given a market regime.

2.3.2. Digging into classification models

Considering each market regime separately, it is also possible to identify a different distribution of outperforming factors. Below is the data for each regime.

Factor	Normal Regime (%)	Correction Regime (%)
Growth	8.37	9.2
Value	15.48	42.53
Momentum	22.18	10.34
Low Volatility	10.88	10.35
Quality	34.73	21.84
Small Cap	8.37	5.75

Table 2.3: Comparison of winning probabilities for each factor

In training the two ML models, the data on the returns of each factor were divided into two parts making use of the labels identified through clustering in the first part. This resulted in two subdatasets, one for each market regime.

The winning probabilities for each factor in the normal and correction regimes generally align with expectations based on typical market behaviors:

- **Value:** The probability is significantly higher in the correction regime. This aligns with the expectation that value stocks tend to outperform during market downturns, as investors seek undervalued assets as a safer option.
- **Growth:** The probabilities remain fairly consistent across both regimes. Growth stocks typically underperform during corrections, so this result suggests some stability, although growth is not a defensive choice.
- **Momentum:** The probability is higher in the normal regime, which fits the typical behavior of momentum strategies, as they perform better in stable or rising markets rather than in volatile correction periods.
- **Low Volatility:** The probability is quite consistent across both regimes, reflecting that low-volatility stocks tend to provide stability and perform similarly in both normal and correction markets.
- **Quality:** The probability remains relatively high in both regimes, though it decreases in the correction regime. Quality stocks are usually considered reliable in both market conditions, but the drop may reflect pressure on profitability during corrections.
- **Small Cap:** The probability decreases in the correction regime, aligning with the expectation that small-cap stocks are more vulnerable to market volatility and perform better in stable conditions.

Model's result for the Normal-market regime

The Random Forest model used in the normal market regime achieves a moderate accuracy of 59%. The confusion matrix shows that the model performs better in some classes than others. Precision, which measures the correctness of positive predictions, varies widely across classes, from 85% for some classes to 29% for others. Comparable results for Recall, F1-score. The macro average, which treats all classes equally, shows moderate overall performance with precision at 0.55, recall at 0.52, and an F1-score of 0.51. The weighted average, which considers the size of each class, shows slightly better results, with precision at 0.64, recall at 0.59, and an F1-score of 0.60.

It must be said that these results can be influenced by the imbalance between classes which is difficult to manage when there is not much data available and when there are 6 classes as in this case. Surely with more focus on these two aspects, better results could have been achieved. It is important to note that the model assesses the correctness in predicting the winning factor (highest probability) and not the individual probabilities for each factor that will actually be used later.

Model's result for the Correction-market regime

If the scarcity of data did not give problems in training the model in the previous case, the same cannot be said for the case of the correction-market regime. Indeed, for one of the 6 factors in the training process, there were not enough samples to resample. It was therefore decided not to consider this class in the model training process, thus making it impossible to predict in the test phase.

The Random Forest model used in the correction market regime presents more challenges compared to the normal regime, reflected by a slight decrease in accuracy to 57%. The confusion matrix and classification report illustrate that the model struggled with class imbalance, which was more pronounced in this case due to the smaller dataset available for training.

The exclusion of one of the factors from the model's training process, making it impossible to predict in the test phase. This decision likely impacted the overall performance of the model, as it limited the model's ability to fully differentiate between all factors.

The overall macro average precision, recall, and F1-score were 0.39, 0.56, and 0.42, respectively. These lower values compared to the normal market regime highlight the model's struggle with imbalanced classes and limited data regime.

Plot of the predicted probabilities: After training the models with the data available up to 2010, it was possible to test them on out-of-sample data by predicting the probabilities for each factor in a given market regime. This was done so that the output data could then be represented in a single graph, which is shown below:

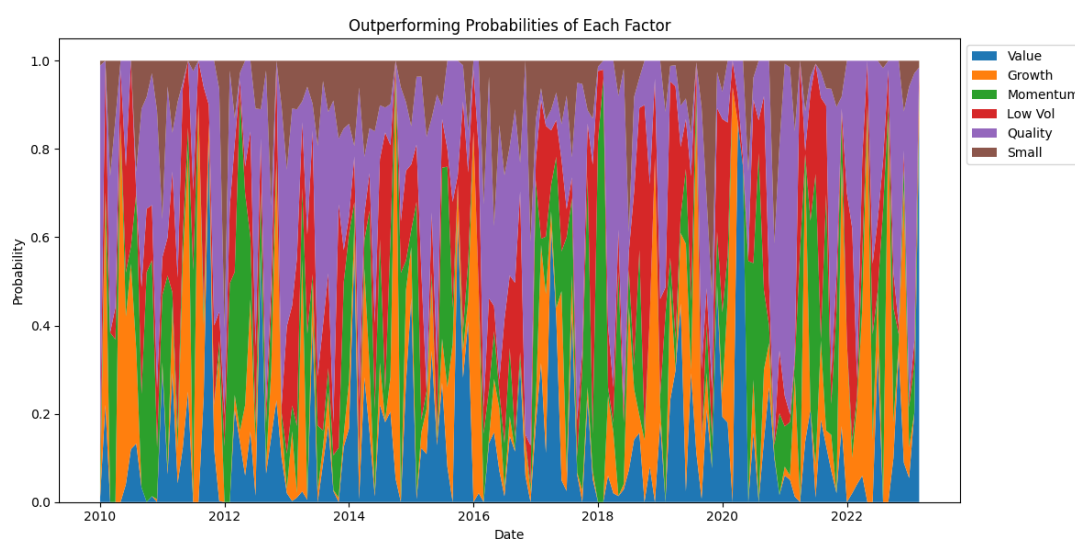


Figure 2.3: Outperforming probabilities of each factor

3 | Strategies results

Arrived at this point with the 3 necessary models available, it is possible to make considerations about the effectiveness of the described ML approach. The metrics using daily data on the SP500, equally weighted strategy, and risk parity shown below will be used as reference.

Strategy	AnnRet (%)	AnnVol (%)	Sharpe	MaxDD (%)	Calmar
S&P 500 Daily EW	9.91	17.06	0.58	-34.85	0.28
Risk Parity Daily	9.65	16.72	0.58	-34.57	0.28
S&P 500 Daily	9.97	17.79	0.56	-33.92	0.29

Table 3.1: Performance metrics sorted by Sharpe ratio

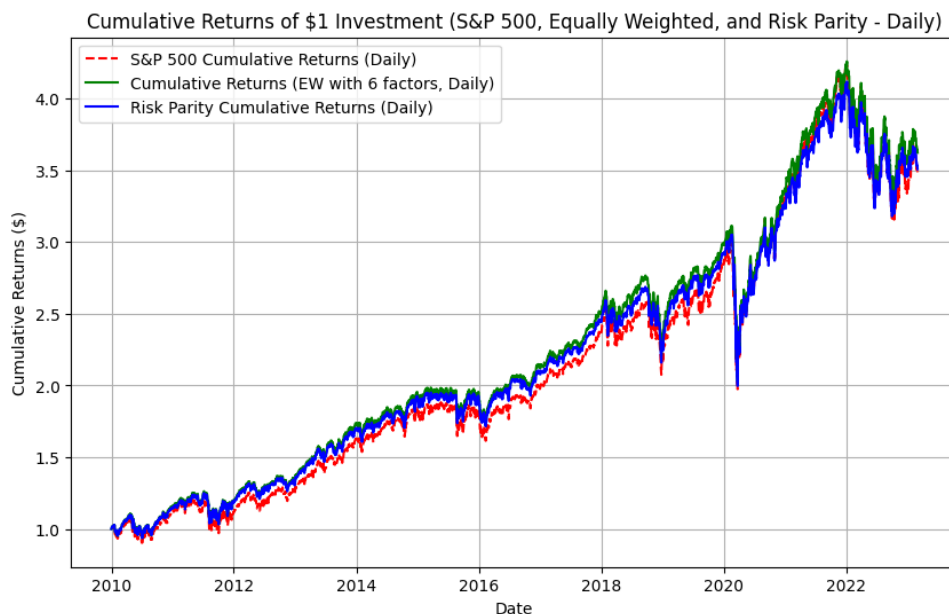


Figure 3.1: Cumulative returns for the S&P 500, Risk Parity, and Equally Weighted

3.1. Comparison between strategies

The next step consisted of testing the previously trained models on the out-of-sample data (2010-2023 time interval) to first predict market regimes, and second the probabilities to be assigned to each factor to weight the portfolio. The daily results will be presented here. Metrics for various strategies were calculated using two weighting schemes:

- **Equally Weighted:** Each factor in the portfolio is assigned an Equally Weighted.
- **Probability-Based Weighted:** Factors are weighted in proportion to their probabilities, with the factor most likely to outperform receiving the highest weight.

In both weighting approaches, the sum of the weights was constrained to equal 1. In the probability-based case, the best-performing factors were selected, and their probabilities were scaled to ensure this requirement was met.

Below are the graphs and the metrics related. The risk parity curve was taken as a reference

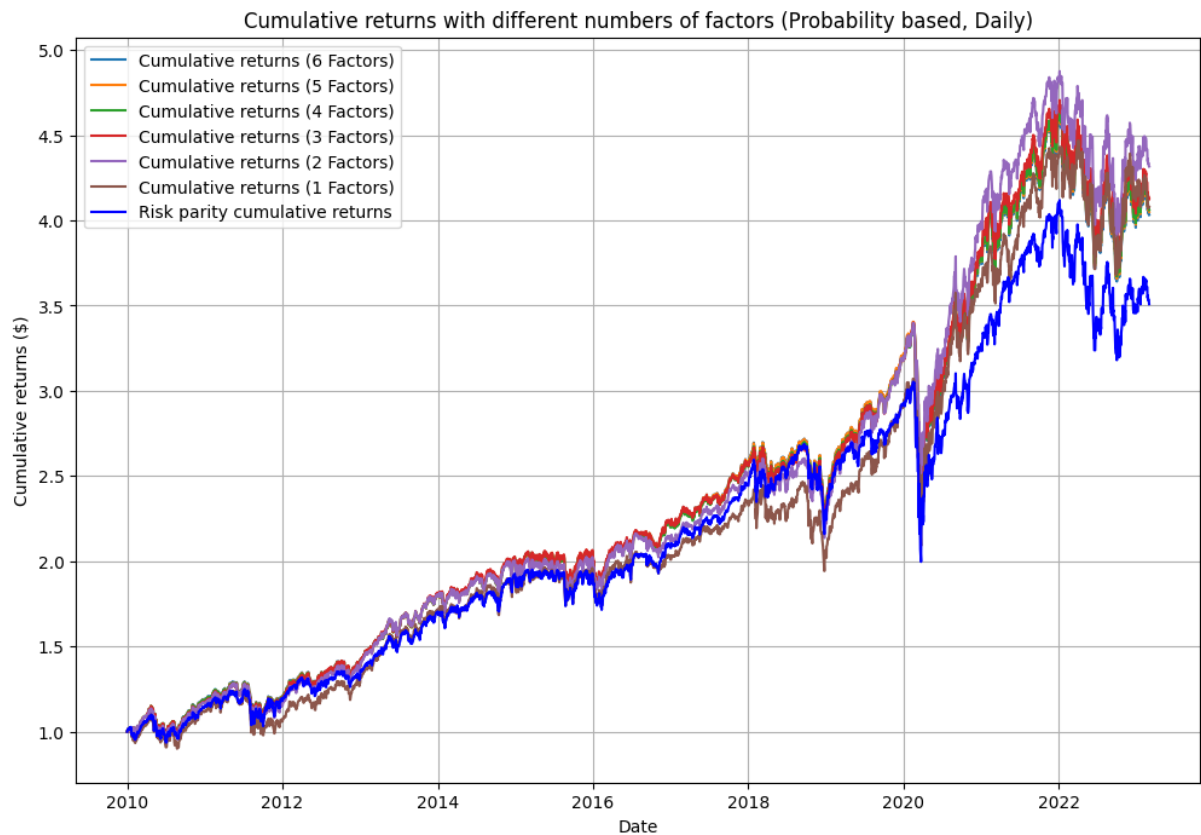


Figure 3.2: Cumulative returns with different numbers of factors using found probabilities 2.3(Daily)

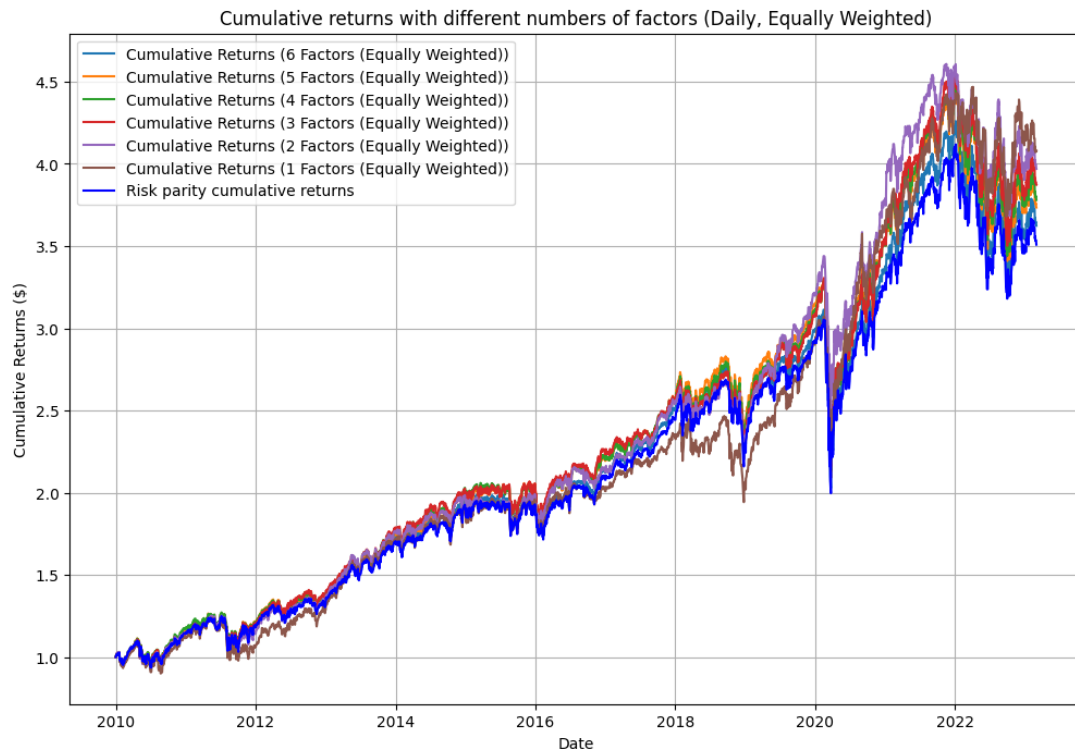


Figure 3.3: Cumulative returns with different numbers of factors using an equally weighted strategy (Daily)

Table 3.2: Performance metrics sorted by Sharpe Ratio

Strategy	AnnRet (%)	AnnVol (%)	Sharpe	MaxDD (%)	Calmar
2 Factors	11.33	17.03	0.67	-31.62	0.36
3 Factors	10.96	17.00	0.64	-33.19	0.32
4 Factors	10.83	16.94	0.64	-34.17	0.32
5 Factors	10.81	16.94	0.64	-34.23	0.31
6 Factors	10.77	16.93	0.64	-34.38	0.31
2 Factors (Equally Weighted)	10.65	16.88	0.63	-31.49	0.34
3 Factors (Equally Weighted)	10.47	16.78	0.62	-34.68	0.30
1 Factor	10.87	17.67	0.61	-31.91	0.34
1 Factor (Equally Weighted)	10.87	17.67	0.61	-31.91	0.34
4 Factors (Equally Weighted)	10.25	16.91	0.61	-35.10	0.31
5 Factors (Equally Weighted)	10.15	16.92	0.60	-35.46	0.29
6 Factors (Equally Weighted)	9.91	17.06	0.58	-34.85	0.28
Risk Parity Daily	9.65	16.72	0.58	-34.57	0.28
S&P 500 Daily	9.97	17.79	0.56	-33.92	0.29

The table above 3.2 provides a detailed breakdown of the performance metrics for various

strategies, including those based on factor investing, equally weighted approaches, Risk Parity, and the S&P 500. A closer analysis of these results reveals interesting insights into the relationship between factor selection, weighting, and overall portfolio performance.

Starting with the number of factors, it is evident that a more focused approach, using fewer factors, tends to yield better outcomes. The 2-factor strategy, for example, stands out with the highest Sharpe ratio of 0.665, suggesting that concentrating on a smaller number of key factors enhances risk-adjusted returns. As the number of factors increases to 6, the Sharpe ratio declines to 0.636, indicating diminishing returns from diversification beyond a certain point. This is further confirmed by the annualized returns, where the 2-factor strategy delivers 11.33%, while the 6-factor strategy drops to 10.77%. Thus, while diversification can help reduce unsystematic risk, over-diversifying by including too many factors seems to dilute the performance benefits. Same thing is seen in the case where only one factor is used, in which case there is an underdiversification of the portfolio.

The volatility (AnnVol) remains relatively stable across the different factor strategies, hovering around 17%. This suggests that the risk, as measured by volatility, is not drastically affected by the number of factors included. However, what is notable is the trend in maximum drawdown (MaxDD). The 2-factor strategy experiences a drawdown of -31.61%, whereas the 6-factor strategy sees a slightly larger drawdown of -34.38%. This implies that while adding more factors may not significantly increase volatility, it can expose the portfolio to deeper drawdowns during market downturns, which is reflected in the Calmar ratio (a measure of return relative to drawdown). The 2-factor strategy again outperforms with a Calmar ratio of 0.358, compared to 0.313 for the 6-factor strategy, emphasizing the advantage of focusing on fewer, stronger factors.

When comparing probability-based versus equally weighted approaches, it is clear that weighting the factors based on their importance or probabilities provides a performance boost. For example, the 2-factor equally weighted strategy has a Sharpe ratio of 0.630, lower than the 0.665 of its probability-based counterpart. Similarly, the annualized return for the equally weighted strategy is 10.65%, compared to 11.33% for the probability-based strategy. This suggests that a more intelligent weighting scheme, which emphasizes the most relevant factors, results in better outcomes than a naive equal-weight approach. Equally weighted strategies generally exhibit slightly lower volatility, but this comes at the cost of reduced returns and slightly weaker risk-adjusted performance.

In comparing the factor-based strategies to Risk Parity, we can observe several key differences in terms of performance, risk, and overall consistency. Risk Parity serves as a reference for a balanced, well-diversified approach designed to allocate risk equally across

various assets or factors. With a Sharpe ratio of 0.577, Risk Parity performs reasonably well, slightly outpacing the S&P 500 in terms of risk-adjusted returns, which has a Sharpe ratio of 0.561. However, it's important to note that despite its balanced approach, Risk Parity's overall performance is not superior to the factor-based strategies. The 2-factor probability-based strategy, for example, significantly outperforms Risk Parity. This suggests that a concentrated exposure to the top two factors leads to much better risk-adjusted returns compared to the broader risk-spreading approach of Risk Parity. The annualized return for the 2-factor strategy also stands out at 11.33%, well above the 9.65% achieved by Risk Parity. This indicates that focusing on a small number of highly relevant factors can deliver superior returns without taking on excessive risk.

Overall, these results highlight several key takeaways. First, focusing on a small number of carefully chosen factors leads to the best risk-adjusted performance, as seen with the 2-factor strategy. Second, applying intelligent weighting to these factors, rather than Equally Weighting, enhances returns and improves Sharpe ratios. These findings align with the broader understanding that diversification and weighting schemes are critical in portfolio construction.

Reported below the two best strategies along with two of their benchmarks, the SP500 and the Risk parity strategy.

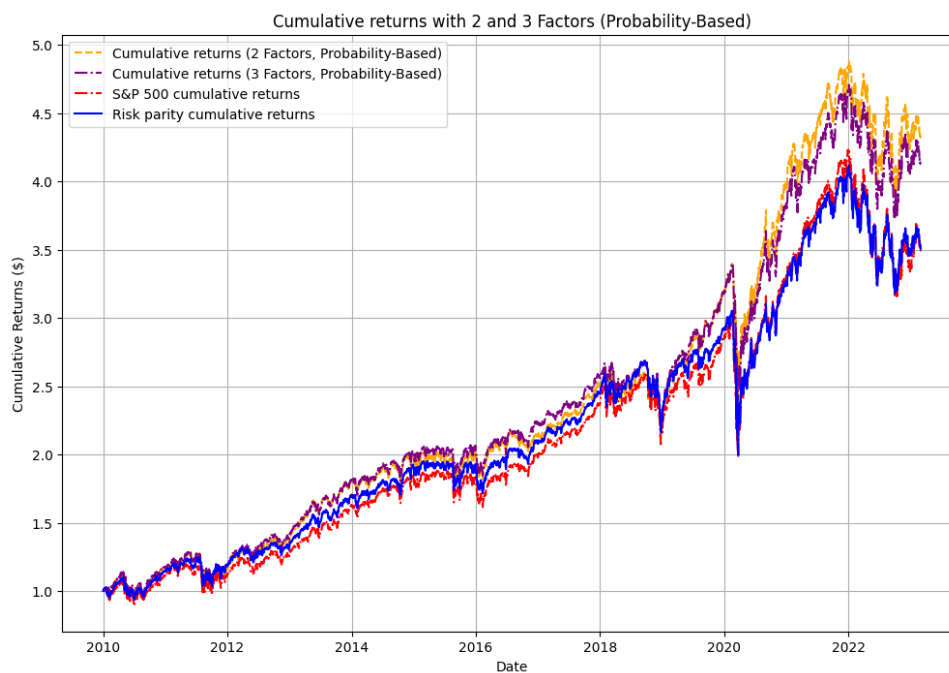


Figure 3.4: Cumulative returns with 2 and 3 Factors (probability-based), S&P 500 and Risk Parity

The chart above 3.4 compares the cumulative returns of the 2-factor probability-based strategy, 3-factor probability-based strategy, S&P 500, and Risk Parity over a period spanning from 2010 to 2023. The visual representation highlights the relative performance of each strategy and reveals some important dynamics between these approaches.

One of the most striking observations is the clear outperformance of the 2 and 3-factor probability-based strategies over both the S&P 500 and Risk Parity, especially in the latter part of the period. Initially, from 2010 to around 2015, all strategies tracked each other quite closely. This is indicative of a stable market environment where both broad market strategies (S&P 500 and Risk Parity) and concentrated factor-based strategies performed similarly. However, after 2019, the factor-based strategies, particularly the 2-factor probability-based strategy, began to diverge, showing increasingly superior performance.

The 2-factor probability-based strategy ultimately achieves the highest cumulative returns, quadrupling the initial investment by the end of the period. This suggests that focusing on a small, carefully selected set of factors allows the strategy to capture more market opportunities. The 3-factor strategy follows a similar but slightly less pronounced trajectory.

Another notable point is how the factor-based strategies handled major market downturns, such as the 2018 correction and the COVID-19 crash. While both the S&P 500 and Risk Parity experienced sharp declines, the 2 and 3-factor strategies demonstrated a quicker recovery and resumed their upward trend more decisively.

3.2. Long-short strategies

Long-short strategies offer advantages compared to the long-only approaches discussed earlier, particularly in terms risk management. If the short position in a long-short strategy is based on a benchmark, such as the S&P 500, the risk parity or in this case the equally weighted on 6 factors, the dynamics of the strategy shift significantly. This structure often transforms the strategy into a market-neutral or benchmark-neutral approach, with several important implications. One key effect of shorting the benchmark is that the strategy becomes focused on outperformance rather than relying on overall market growth. The overall market direction becomes less important, and the strategy's success hinges on the ability to identify and invest in assets or factors that consistently outperform the market. This makes the strategy particularly useful in bear markets or during periods of market uncertainty when long-only strategies might struggle. In this setup, the performance of the portfolio is no longer dictated by the market's overall performance

but by the relative strength of the long positions compared to the benchmark.

By shorting the benchmark, the strategy also reduces its systematic risk—the risk tied to the overall market. Instead of depending on the broader market to rise, the strategy isolates and concentrates on idiosyncratic risks, which are specific to the assets or factors held in the long positions.

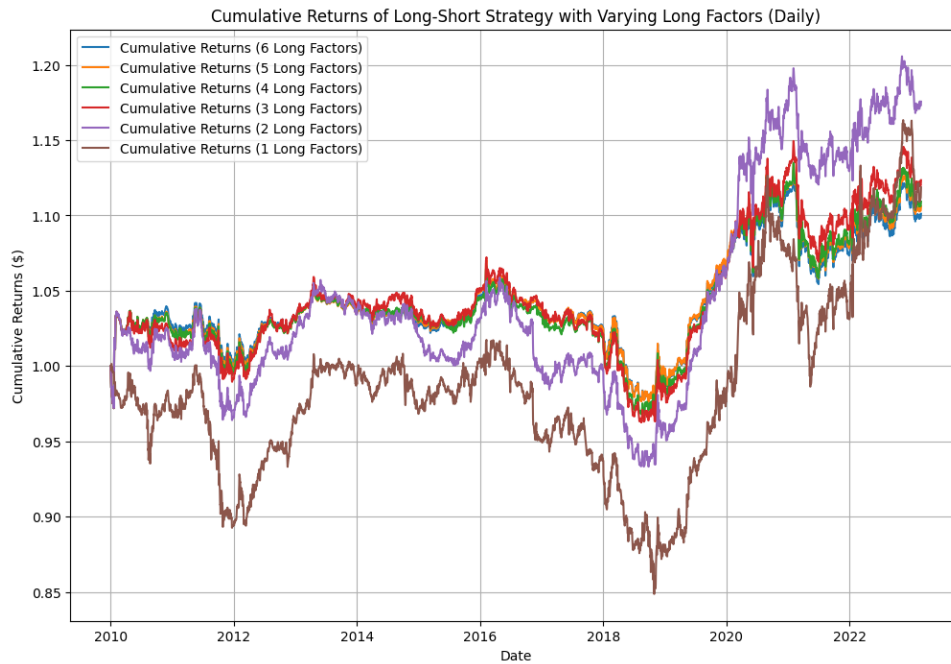


Figure 3.5: Cumulative returns of the strategy with varying # of probability-based long factors and short 6 EW factors

The graph 3.5 illustrates the cumulative returns of a long-short strategy, where the long positions are taken on a varying number of weighted factors (ranging from 1 to 6), and the short position is held on an equally weighted basket of all six factors. The results show that the strategies with more long-factors (4 to 6 factors) maintain relatively stable cumulative returns throughout the period, staying close to 1.0, which indicates consistent but unremarkable performance. The 1-factor strategy experiences greater volatility and more severe drawdowns, especially during periods of market turmoil. On the other hand, the 2-factor and 3-factor strategies emerge as the top performers in the long-short setup. These strategies, particularly after 2019, show superior cumulative returns, reaching values around 1.1 by the end of the period. However, their drawdowns during volatile periods indicate that the model's factor selection process is not fully adaptive to abrupt shifts in downward market.

4 | Conclusion

The aim of this project was to evaluate the performance of an equity factor timing approach, which incorporates market regimes and utilizes machine learning algorithms, in comparison to traditional investment methodologies.

The analysis revealed that the considered factor method consistently delivers better performances compared to the others approaches. However, while from a graphical and theoretical point of view this is true, it is also true that from a practical point of view at present these results hardly hint at an effective application of the methodology. This project is intended to be a starting point, the basis for future improvements.

It was observed that during periods of high volatility, such as the 2020 COVID-19 crisis, the model had difficulty adapting to the rapid changes in factor dynamics. It is also important to note that transaction costs were not accounted for in this analysis. Looking ahead, further improvements are likely needed, particularly in the machine learning training process.

In particular we can state that:

- The model that identifies market regimes is performing well. As illustrated in the plot below 4.1, the strategies developed using the predicted regimes and those using the actual regimes are very similar. This similarity indicates that the model is effectively capturing the regime shifts.
- The model tasked with identifying the best-performing factor could benefit from further enhancements. We suggest expanding the feature set to include beyond alternative data sources (i.e news sentiment indicators, etc ..) also additional technical indicators, such as:
 - **MACD indicators** (Moving Average Convergence Divergence)
 - **Volatility and drawdown indicators**
 - **Relative Strength Index (RSI)**

Incorporating these elements would allow the model to capture additional dimensions of asset performance, improving its predictive capabilities.

- Further tests could also be conducted, exploring:
 - different training methodologies (e.g., *rolling* or *expanding window*),
 - alternative algorithms beyond *Random Forest (RF)*, such as *XGBoost*, *neural networks*, and *learning-to-rank* algorithms designed specifically to tackle asset ranking problems.
- Additional target variables may also be worth exploring. Instead of defining a single class for the top-performing factor, we could categorize factors based on different criteria.

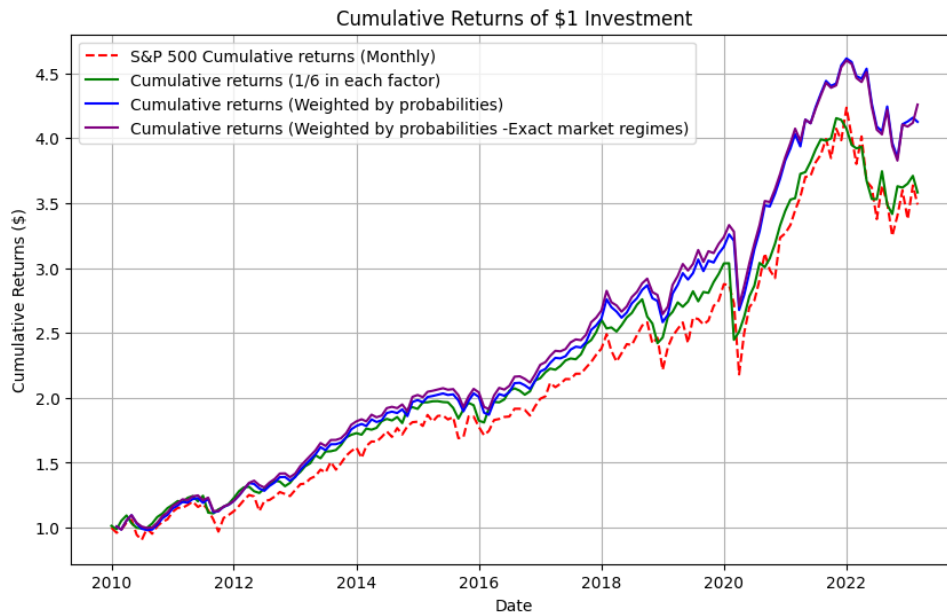


Figure 4.1: Cumulative returns of \$1 investment (S&P 500 Monthly, 1/N Factors, probability-based factors using predicted and exact market-regime labels)

In addition, as mentioned in the previous chapters, using the Russell as a benchmark index could lead to improvements by giving a broader view of the market.

In summary, while the findings presented provide a strong foundation for understanding the potential of machine learning and factor-based strategies in portfolio management, there remains significant room for refinement. By continuously improving these strategies, they can become viable tools for real-world application, providing investors with robust, adaptive portfolios capable of navigating both stable and turbulent market environments.

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A | Appendix A

Legend for table sectors:

Inflation	Production & Business Activity	Financial Conditions	
Monetary Policies	Equity Earning Yield	Labor	National Accounts

Variable	Definition	Description
CPILFESL	Consumer Price Index for All Urban Consumers: All Items Less Food and Energy in U.S. City Average	A measure of inflation excluding volatile food and energy prices.
CPILFESL1YAC	CPILFESL 1-Year Annualized Change (%)	Annual percentage change in the CPILFESL.
CPILFESL10YAC	CPILFESL 10-Years Annualized Change (%)	10-year percentage change in the CPILFESL.
CPIAUCSL	Consumer Price Index for All Urban Consumers: All Items in U.S. City Average	Measures inflation in the overall economy.
CORESTICKM159SFRE	Sticky Price Consumer Price Index less Food and Energy	Measures inflation for items with prices that change infrequently.
PCEPI	Personal Consumption Expenditures: Chain-type Price Index	Reflects changes in consumer prices.
PCEPILFE	Personal Consumption Expenditures Excluding Food and Energy (Chain-Type Price Index)	Measures inflation excluding food and energy for personal consumption.
MEDCPIM158SFRBCLI	Median Consumer Price Index	A measure of central inflation trends.
PPIACO	Producer Price Index by Commodity: All Commodities	Measures inflation from the perspective of sellers.
PCUOMFGOMFG	Producer Price Index by Industry: Total Manufacturing Industries	Price changes in the manufacturing sector.
ISRATIO	Total Business: Inventories to Sales Ratio	Reflects how efficiently inventories are managed.
BUSINV	Total Business Inventories (Millions of Dollars)	Total value of inventories in businesses.
TTLCONS	Total Construction Spending: Total Construction in the United States (Millions of Dollars)	Measures the total spending on construction in the U.S.
TLMFGCONS	Total Construction Spending: Manufacturing in the United States (Millions of Dollars)	Total spending on manufacturing-related construction.
USSTHPI	All-Transactions House Price Index for the United States	Measures changes in house prices across the U.S.
PERMIT	New Privately-Owned Housing Units Authorized in Permit-Issuing Places: Total Units (Thousands of Units)	Reflects the number of new housing units authorized for construction.
INDPRO	Industrial Production: Total Index	Measures the output of the industrial sector.
TCU	Capacity Utilization: Total Index	Measures the extent to which industrial capacity is used.

IPMAN	Industrial Production: Manufacturing (NAICS)	Industrial production in the manufacturing sector.
MNFCTRIRSA	Manufacturers: Inventories to Sales Ratio	Inventory levels compared to sales in manufacturing.
WHLSLRIRSA	Merchant Wholesalers: Inventories to Sales Ratio	Inventory levels compared to sales in wholesale trade.
USSLIND	Leading Index for the United States (%)	A leading indicator of future economic activity.
USPHCI	Coincident Economic Activity Index for the United States	Measures current economic conditions.
JHGDPRINDX	GDP-Based Recession Indicator Index (%)	Predicts the probability of a recession based on GDP data.
GDPC1	Real Gross Domestic Product (Billions of Chained 2017 Dollars)	Measures the inflation-adjusted value of goods and services produced by the economy.
NFCIRISK	Chicago Fed National Financial Conditions Risk Subindex	Measures risk in the U.S. financial system.
NFCICREDIT	Chicago Fed National Financial Conditions Credit Subindex	Measures credit conditions in the U.S. financial system.
NFCILEVERAGE	Chicago Fed National Financial Conditions Leverage Subindex	Measures leverage conditions in the U.S. financial system.
DFE	Federal Funds Effective Rate (%)	The interest rate at which banks lend to each other.
M1REAL	Real M1 Money Stock (Billions of 1982-84 Dollars)	The real value of the narrow money supply.
M2REAL	Real M2 Money Stock (Billions of 1982-84 Dollars)	The real value of the broad money supply.
T10YFF	10-Year Treasury Constant Maturity Minus Federal Funds Rate (%)	The difference between long-term and short-term interest rates.
CAPE	Cyclically Adjusted PE Ratio	A long-term measure of equity market valuation.
ICSA	Initial Claims (Number of Claims)	Measures the number of new claims for unemployment benefits.
IURSA	Insured Unemployment Rate (%)	The percentage of the labor force receiving unemployment insurance.
JTSJOL	Job Openings: Total Nonfarm (Thousands)	Measures the number of available job openings.
PAYEMS	All Employees, Total Nonfarm (Thousands)	Measures the total number of employees on nonfarm payrolls.
MANEMP	All Employees, Manufacturing (Thousands)	Measures the number of employees in manufacturing.
UNRATE	Unemployment Rate (%)	The percentage of the labor force that is unemployed.
IEAXGS	Exports of goods and services (Millions of Dollars)	Measures the total value of goods and services exported.
IEAMGSN	Imports of goods and services (Millions of Dollars)	Measures the total value of goods and services imported.
GFDEGDQ188S	Federal Debt: Total Public Debt as Percent of GDP (%)	Measures the total public debt as a percentage of GDP.

Table A.1: Summary of macroeconomic variables organized by sector

B | Appendix B: Features selection

The results obtained were considering all features. However, it could be analyzed how these vary if the number of features is reduced. Two methods for reducing the number of features have been implemented in the code: Random Forest selection and Mutual Information. Due to lack of time, no attempt has been made, but it might be something to analyze later if this project is taken as a reference in the future.

In the table below B.1, the features that contribute more than 3% to the RF's predictions have been identified. A threshold of 3% was chosen to focus on the most relevant features, highlighting their impact on the model's performance. These features are crucial as they hold the most predictive power and are therefore prioritized in the analysis.

Feature	Importance (%)
CAPE	20.04
NFCICREDIT	14.77
Financial turbulence	11.41
NFCIRISK	10.89
MANEMP	6.57
NFCILEVERAGE	4.59
TCU	4.02
UNRATE	3.92
USPHCI	3.70
PAYEMS	3.56
CORESTICKM159SFRBATL	3.34

Table B.1: Features contributing more than 3% to the RF's predictions.

Feature	Mutual Information Score
NFCIRISK	0.190585
CAPE	0.183786
NFCICREDIT	0.177725
NFCILEVERAGE	0.173185
Financial turbulence	0.139541
JHGDPBRINDX	0.127286
PAYEMS	0.108501
MANEMP	0.106873
MEDCPIM158SFRBCLE	0.100433
UNRATE	0.086170
USPHCI	0.080610
DFF	0.072491
IPMAN	0.064371
TCU	0.059770
ICSA	0.053079
M2REAL	0.037550
PPIACO	0.035166
USSTHPI	0.035095

Table B.2: Features with a Mutual Information score greater than 0.03.

Instead, in B.2, features with a Mutual Information (MI) score greater than 0.03 were selected. The mutual information score measures the dependency between features and the target variable, identifying which features provide the most information for prediction. Mutual Information (MI) and Random Forest (RF) feature importance both serve as methods for feature selection, but they have different strengths and weaknesses. MI measures the dependency between each feature and the target variable, making it particularly useful for detecting non-linear relationships that may not be easily captured by other methods. However, MI considers each feature independently, meaning it does not account for interactions between features or their combined impact on the target variable. This can limit its effectiveness in models where feature interactions play a significant role.

C | Appendix C: Overview of classification model parameters

1. Random Forest

Random Forest is an ensemble learning method that constructs multiple decision trees during training and outputs the class that is the mode of the classes predicted by the individual trees. It improves model accuracy and helps avoid overfitting by averaging over several trees.

The parameter grid for Random Forest includes the following hyperparameters:

- **smote_sampling_strategy**: Controls the ratio of the number of samples in the minority class to the number of samples in the majority class after applying SMOTE. Values range from 0.6 to 1.0.
- **smote_k_neighbors**: Specifies the number of nearest neighbors used to generate synthetic samples for the minority class.
- **classifier_n_estimators**: The number of trees in the forest. More trees generally lead to better performance, but increase computation time. Values range from 100 to 600.
- **classifier_max_depth**: The maximum depth of each tree. A larger depth allows the model to learn more complex patterns but may cause overfitting.
- **classifier_min_samples_split**: The minimum number of samples required to split an internal node. Higher values prevent overfitting by controlling the complexity of the tree.
- **classifier_min_samples_leaf**: The minimum number of samples required to be at a leaf node. Higher values can smooth the model and reduce overfitting.

2. SVC

Support Vector Classifier is a linear model for classification that works by finding the hyperplane that best separates the classes in the feature space. It can also be extended to non-linear classification by using kernel functions.

The parameter grid for SVC includes the following hyperparameters:

- **smote_sampling_strategy** and **smote_k_neighbors**: Same as for the previous model
- **classifier_C**: Regularization parameter that controls the trade-off between achieving a low training error and a low testing error. Larger values of C reduce the regularization strength.
- **classifier_kernel**: Specifies the kernel type to be used in the algorithm. Options include linear or polynomial kernels.
- **classifier_gamma**: Defines how far the influence of a single training example reaches. Higher values of gamma lead to a model that fits the training data closely but may overfit.

3. GNB

Gaussian Naive Bayes (GNB) is a probabilistic classifier based on Bayes' theorem. It assumes that the features follow a Gaussian (normal) distribution and that the features are independent given the class.

The parameter grid for Gaussian Naive Bayes includes the following hyperparameters:

- **smote_sampling_strategy** and **smote_k_neighbors**: Same as for the previous models
- **classifier_var_smoothing**: Adds a small value to the variance to prevent numerical errors during computation, especially when the variance is close to zero. Smaller values lead to a more detailed model.

4. Stacking classification approach

In the binary classification the outputs of the base classifiers continue to be used as inputs for the **Logistic Regression (LR)** classifier with the difference that it is trained to learn only 3 optimal weights, w_1 , w_2 , and w_3 . The loss function is defined as:

$$L(w) = - \sum_i [y_i \log(y_{\text{pred}_i}) + (1 - y_i) \log(1 - y_{\text{pred}_i})]$$

where y_i is the true label of the i -th sample, and y_{pred_i} is the predicted probability for the i -th sample obtained from the stacked ensemble.

The final prediction y_{pred} for a given input x is computed as follows:

$$y_{\text{pred}} = \sigma(w_1 \cdot RF(x) + w_2 \cdot SVC(x) + w_3 \cdot GNB(x))$$

where σ is the sigmoid function that maps the weighted sum of the base classifiers' outputs to a probability between 0 and 1. Each model will have as output 0 or 1 depending on the aforementioned class. This allows the logistic regression model to adapt to the underlying data and improve overall classification performance.

List of Figures

1.1	S&P500 monthly drawdown	4
1.2	Elbow method visualization	6
1.3	Clustering and drawdown visualization	7
1.4	Financial turbulence	9
2.1	Global correlation matrix	17
2.2	Normal market correlation matrix (left) and Correction market correlation matrix (right)	18
2.3	Outperforming probabilities of each factor	21
3.1	Cumulative returns for the S&P 500, Risk Parity, and Equally Weighted .	22
3.2	Cumulative returns with different numbers of factors using found probabilities	
2.3(Daily)		23
3.3	Cumulative returns with different numbers of factors using an equally weighted strategy (Daily)	24
3.4	Cumulative returns with 2 and 3 Factors (probability-based), S&P 500 and Risk Parity	26
3.5	Cumulative returns of the strategy with varying # of probability-based long factors and short 6 EW factors	28
4.1	Cumulative returns of \$1 investment (S&P 500 Monthly, 1/N Factors, probability-based factors using predicted and exact market-regime labels)	30

List of Tables

1.1	Comparison of market regime clustering results	6
1.2	Test results for the RF model	14
1.3	Test results for the SCA	14
2.1	Selected equity factors based on the S&P 500 Index	16
2.2	Winning probabilities for each factor	19
2.3	Comparison of winning probabilities for each factor	19
3.1	Performance metrics sorted by Sharpe ratio	22
3.2	Performance metrics sorted by Sharpe Ratio	24
A.1	Summary of macroeconomic variables organized by sector	33
B.1	Features contributing more than 3% to the RF's predictions.	34
B.2	Features with a Mutual Information score greater than 0.03.	35